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The Wall Friction Properties of Pharmaceutical Powders, Blends, and Granulations

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ABSTRACT

Data from wall friction testing and physical property characterization of over 100 pharmaceutical powders, blends, and granulations have been analyzed. The analyses focused on data for stainless steel surfaces with the most common finishes for pharmaceutical powder processing equipment, either a 2B cold rolled mill finish or an electropolished 2B surface. Active pharmaceutical ingredients exhibited the highest friction against these surfaces, whereas active granulations exhibited the least friction. The typical (median) wall friction angle for an active blend on 2B stainless steel was 22° versus 18° for an active granulation. Typical wall friction values on electropolished 2B surfaces were about 17° and 12° for active blends and granulations, respectively. Blends typically exhibited larger wall friction angles than the granulations suggesting that simple blends will usually require hoppers or bins with steeper walls to achieve mass flow. Lower wall friction angles were consistently observed against the smoother electropolished 2B surface, and, thus, the wall surface finish should be considered when designing bins and hoppers for use with pharmaceutical powders. The wall friction angles of blends and granulations did not show any definite trend as the percentage of active pharmaceutical ingredient increased.

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Introduction

The flow of pharmaceutical powders from hoppers, bins, and chutes is determined in part by the friction of the powder with the surface of the equipment. Therefore, the so-called “wall friction” of such materials is measured routinely so that estimations of powder flow performance in industrial processes can be made.^{1–4} For a variety of reasons, data of this type have very rarely been reported in the scientific literature and typical wall friction values for pharmaceutical powders are not well known. In addition, the influences (if any) of prior processing history or particle properties are largely unknown.

Background

The data needed to predict the flow mode (i.e., mass flow or funnel flow) of a given powder in a given hopper or bin are the

hopper wall angle, the powder's effective angle of internal friction, and the powder's wall friction angle (WFA). The hopper wall angle, θ , is the slope in degrees of the hopper wall measured from vertical. The powder's effective angle of internal friction, δ , is a measure of the friction within the powder when sheared internally and is expressed in degrees. It can be easily determined from the powder yield locus obtained from shear cell testing (shear cell methodology for powder flow testing, United States Pharmacopeia/National Formulary General Chapter <1063>). The powder's WFA, ϕ' , is a measure of the sliding friction at the powder/wall interface. It is often determined by shear cell testing where the powder under load slides across a coupon representing the hopper wall surface, and it is also expressed in degrees. The principle of the wall friction test is illustrated in Figure 1, where a normal stress, σ_w , is applied that acts between the bulk powder and the wall material. The shear stress τ_w is then measured as the powder slides across the wall sample at velocity v . The process is repeated over a range of normal stresses. A wall yield locus plot, τ_w versus σ_w , is used to analyze the stresses recorded during the test. A typical wall yield locus is shown in Figure 2. The WFA at a given normal stress, ϕ_x , is the slope of the yield locus at that point. In the example in Figure 2, the slope or WFA decreases with increasing normal wall stress. In other instances, the wall yield locus may be a straight line such that the WFA is approximately constant over the range of normal wall stresses considered.

Statements: The primary data used in preparation of the manuscript is available in Tables 1–7. Data for specific materials can be requested from the author at bruno.c.hancock@pfizer.com.

The author is a salaried employee and shareholder of Pfizer Inc. and declares no conflicts of interest (financial or non-financial).

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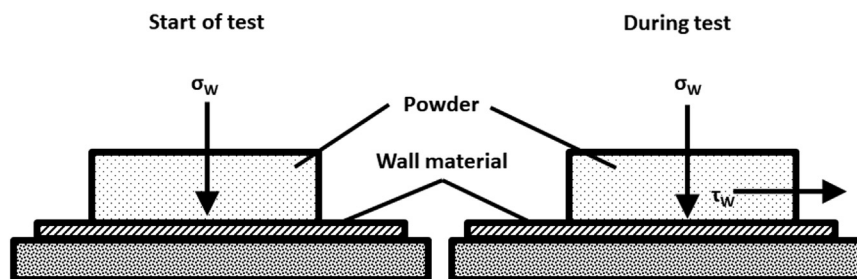


Figure 1. Wall friction test schematic (σ_w = normal stress; τ_w = shear stress).

Knowledge of WFAs is of interest to many industries, including the mining, agricultural, food, and pharmaceutical industries. However, surprisingly little data are found in the public domain. Examples where agricultural grains were evaluated include the work by Ross et al.⁵ on winter wheat and LoCurto⁶ on soybeans. Wall friction data reported for other bulk solids include tetrapotassium pyrophosphate, gypsum and silicon carbide,¹ and fine milled limestone.⁷

Wall friction data for pharmaceutical materials were first reported by Jolliffe and Newton⁸⁻¹¹ in the early 1980s. They studied wall friction effects in capsule filling processes for lactose samples of different particle sizes. The WFAs against steel surfaces with different finishes varied between 8° and 27°, and, in general, the wall friction increased with increasing surface roughness and decreasing particle size. Subsequently, Baichwal and Augsburger^{12,13} measured the coefficient of wall friction for several pharmaceutical lubricants (including magnesium stearate) against a polished steel substrate. The coefficients ranged from 0.099 to 0.312, corresponding to WFAs from 6° to 17° (the coefficient of wall friction is the tangent of the WFA). Such low wall friction values are to be expected for lubricants since their function is to reduce friction between powder and machine surfaces. Later, Tan and Newton^{14,15} studied wall friction effects in their work on automated capsule filling machines and they reported angles of between 11° and 35° for a range of common excipients. There was also a weak trend of greater wall friction with an increasing surface roughness of the equipment and decreasing particle size of the powders in this work. More recently, Freeman et al.¹⁶ reported WFAs from 27.5° to 31.5° for Respitose ML001, a lactose excipient for inhalation. The wall friction testing was performed on 316L stainless steel substrates with a range of surface finishes and the

wall friction values increased with increasing surface roughness. Not long afterward, McCarron et al.¹⁷ showed trends for several different types of pharmaceutical powders, with active pharmaceutical ingredients (APIs; $\phi' > 30$) being more frictional than lubricated blends ($\phi' < 10$). Faulhammer et al.¹⁸ re-visited wall friction effects in encapsulation processes in 2014 and they again reported a weak correlation between particle size and the WFA for microcrystalline cellulose samples. Finally, Van Snick et al.¹⁹ studied wall friction as part of a multivariate study of blend properties that could potentially impact die filling during tabletting. They reported wall friction data for 30 active blends (with values between 4° and 18°) and noted that the lubricated blends exhibited lower friction values than unlubricated blends. These authors also noted positive correlations between wall friction values and both tablet ejection forces and tablet weight variability data that had not been reported previously.

In this article, we report an analysis of wall friction data for over 100 different powder samples used in the manufacture of solid oral dosage forms. The samples were mostly taken from commercial scale industrial manufacturing facilities, although a significant proportion (about 20%) was produced at a smaller scale in the laboratory. To the best of our knowledge, this article represents the first comprehensive compilation of wall friction data for pharmaceutical materials. The range of properties measured and any trends identified in these data should help practicing pharmaceutical scientists in several ways. First, it should enable them to improve the design of the bulk powder handling equipment (such as hoppers and chutes) used with these materials. Second, it should help them to modify the properties of their formulations so that they can avoid material handling problems (such as funnel flow, rat holing, and segregation). Finally, it should contribute to the

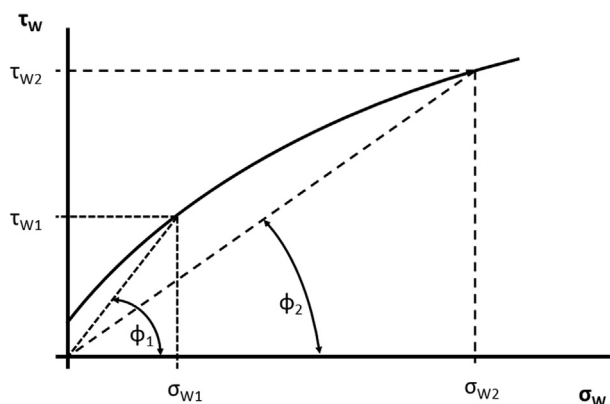


Figure 2. Typical wall yield locus. In this example, the wall friction angle (ϕ) is dependent on the wall normal stress (σ_w).

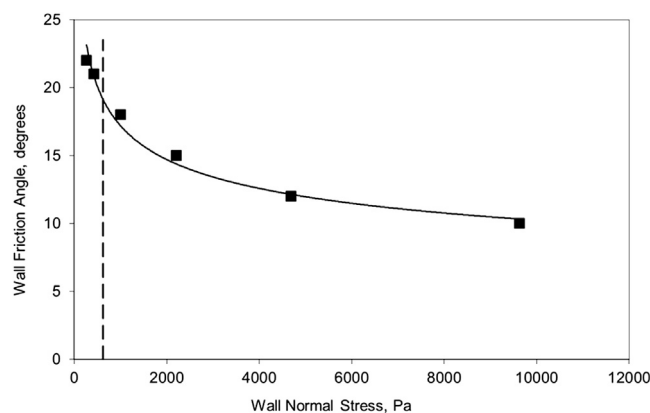


Figure 3. Relationship between wall friction angle and wall normal stress for a typical active blend. Dashed line indicates standard normal stress of 622 Pa. In this example, wall friction angle at the standard stress is 19.1°.

Table 1
Summary Statistics for Physical Properties of all Materials

Variable	N	10th	50th	90th	Minimum	Maximum	Mean
Bulk density (g/cm ³)	133	0.29	0.54	0.69	0.10	1.57	0.53
Tapped density (g/cm ³)	58	0.38	0.72	0.84	0.12	1.05	0.67
Particle size (10th percentile) (μm)	77	7	18	70	2	357	36
Particle size (50th percentile) (μm)	77	57	115	393	6	754	170
Particle size (90th percentile) (μm)	77	179	417	1058	13	1398	512
Span (Bmid80)	77	1.7	2.6	6.3	0.7	13.1	3.5
Wall friction angle, 622 Pa, 2B finish (°)	112	12.3	22.2	32.3	6.0	39.6	22.1
Wall friction angle, 622 Pa, 2B-EP finish (°)	100	6.9	16.7	30.3	3.6	44.7	17.7

10th, 50th, and 90th are percentiles of the population. The wall friction angle at 622 Pa was determined by interpolation.

attainment of increased product quality and improved process efficiency by minimizing material losses and process failures during API and drug product manufacturing operations (e.g., due to the incomplete discharge of hoppers and bins).

Materials and Methods

Wall friction and physical property data were compiled for pharmaceutical powder samples tested for Pfizer Inc. over a 24-year period by Jenike and Johanson Inc. (Tyngsborough, MA). The samples included APIs, excipients, blends, granulations (both wet and dry), and a small number of miscellaneous pharmaceutical powders. The majority of samples were taken from commercial scale lots, and the remainder (about 20%) were taken from laboratory scale trials. Most of the blends and granulations had been lubricated prior to their ultimate manufacture into capsules or tablets.

The wall friction and physical property data included WFAs measured at multiple normal stresses, bulk and tapped densities, and particle size distributions. The wall friction tests were all conducted using a standard Jenike translational shear cell using American Society for Testing and Materials method D6128.^{1,20} Samples were tested “as received” and were kept sealed in their original containers prior to and in between tests. Replicate samples were tested in quick succession to minimize any test-to-test variation in sample water content. The laboratory humidity conditions were monitored during testing and typically were between 40% and 60% relative humidity. The powder samples were consolidated at the highest normal stress prior to the measurement and all tests were made across the grain of the surface finish. The wall friction measurements were performed on 304 or 316 stainless steel coupons with 2B cold-rolled or 2B electropolished (2B-EP) finishes which are representative of pharmaceutical powder processing equipment. Typical roughness (Ra) values for the 2B finish are 0.2–0.6 micron, whereas for the electropolished 2B finish the roughness values are usually between 0.1 and 0.3 micron.²¹

Each experimentally determined wall yield locus (Fig. 2) was transformed into a plot of normal stress versus WFA (Fig. 3). This permitted the determination of WFA at a standard stress (622.4 Pa or 13 lb/ft²) by interpolation. This stress condition was selected for comparison of various powder samples because it is typical of that at the outlet of a hopper or bin used in the manufacture of solid oral dosage forms (tablets and capsules).

Summary statistics were generated for the entire data set and for various data sub-sets (APIs, blends, granulations, etc.), and examined for trends. The data for active blends and granulations were also analyzed by API loading where the API loading ranges were defined as low (<10%), medium (10% to <30%), and high (≥30%). Advanced statistical approaches (such as partial least squares) were applied to the data in an attempt to identify any possible correlations between various properties of the samples.

Results and Discussion

Over the 24-year period considered, wall friction and physical property data were generated for over 100 unique samples (Tables 1 and 2). Physical property measurements (such as tapped density and particle size measurements) were not always conducted, so the number of physical property data points (“N”) differed slightly from the number of wall friction data points. There was no obvious systematic bias to which samples were fully tested and the number of data points available for each property was considered to be large enough to allow meaningful statistical analyses to be performed.

The 100+ samples comprised a wide variety of different pharmaceutical material types, and they were split into 8 different categories for further analysis (Table 1). There were 28 samples whose material type could not be definitively determined (labeled as “unknown”); however, it is thought that most of these were active blends or active granulations (based on their physical properties, manufacturing locations, and lot numbers).

All Materials Analysis

When considered as a whole, the physical properties of the samples covered some very wide ranges (Table 1). This is likely a reflection of the broad range of material types and the large number of different API samples tested. The range of bulk and tapped densities was consistent with that previously reported in the literature,²² and the particle size statistics were typical of materials commonly used in solid dosage form development.^{23,24} Probably the most notable finding of the “all materials” analysis was the extremely wide range of WFAs observed for both the 2B and the 2B-EP stainless steel finishes. In addition, the WFAs on the electropolished surfaces were generally less than on the 2B surfaces and this suggests that smoother surfaces can promote increased flowability in hoppers and bins. Twenty-three materials (a significant percentage) showed the opposite trend (greater wall friction

Table 2
Summary of the Material Types With Wall Friction Data

Material Type	Approximate Percentage of Samples
Active blend	21
Active granulation	30
Placebo blend	3
Placebo granulation	2
Excipient	10
API	8
Microspheres and dispersions	5
Unknown	21

Excipients include microcrystalline cellulose, sorbitol, magnesium oxide, sucrose, calcium carbonate, hypromellose acetate succinate, lactose monohydrate, and croscarmellose sodium.

Table 3
Properties of All the Samples Grouped According to the “Material Type”

Variable	Active Blends				Active Granulations				Placebo Blends				Placebo Granulations			
	N	10th	50th	90th	N	10th	50th	90th	N	10th	50th	90th	N	10th	50th	90th
Bulk density (g/cm ³)	28	0.45	0.55	0.66	40	0.45	0.57	0.64	4	0.23	0.56	0.75	2	0.56	0.60	0.63
Tapped density (g/cm ³)	13	0.72	0.79	0.82	21	0.58	0.71	0.85	2	0.20	0.52	0.84	1	0.64	0.64	0.64
Particle size (10th percentile) (μm)	19	7.4	17.3	29.7	29	7.4	20.9	68.8	2	10.7	11.2	11.6	2	61.3	101.1	140.9
Particle size (50th percentile) (μm)	19	58.9	130.0	206.0	29	63.7	111.1	391.8	2	45.3	103.9	162.4	2	192.2	317.2	442.2
Particle size (90th percentile) (μm)	19	180.0	396.0	671.2	29	278.5	594.0	1108	2	129.3	309.6	489.9	2	472.1	792.5	1113
Span (Bmid80)	19	1.7	2.7	4.0	29	2.1	3.3	10.5	2	2.4	2.7	2.9	2	2.1	2.2	2.2
Wall friction angle 2B (622 Pa) (°)	25	18.5	21.7	25.8	35	10.3	17.9	25.1	4	15.2	25.8	36.1	2	10.1	14.2	18.4
Wall friction angle 2B-EP (622 Pa) (°)	27	9.2	17.1	25.8	36	6.0	11.6	22.2	2	34.0	37.1	40.1	2	7.1	10.1	13.0
Variable	API				Excipients				Microspheres and Dispersions				Unknown			
	N	10th	50th	90th	N	10th	50th	90th	N	10th	50th	90th	N	10th	50th	90th
Bulk density (g/cm ³)	10	0.21	0.46	0.55	13	0.24	0.48	0.66	7	0.23	0.27	0.40	28	0.45	0.57	0.76
Tapped density (g/cm ³)	5	0.54	0.67	0.78	5	0.34	0.35	0.76	6	0.37	0.39	0.52	5	0.66	0.72	0.80
Particle size (10th percentile) (μm)	5	5.9	9.9	38.8	7	9.6	36.8	333.6	3	19.8	27.0	118.2	10	9.8	17.4	37.8
Particle size (50th percentile) (μm)	5	65.6	129.0	412.0	7	30.4	108.3	609.4	3	61.0	69.0	174.6	10	63.7	127.7	381.5
Particle size (90th percentile) (μm)	5	348.8	535.6	983.2	7	74.5	194.5	1047	3	141.8	153.0	258.6	10	201.9	413.1	702.2
Span (Bmid80)	5	2.3	3.3	6.7	7	1.1	1.7	2.2	3	0.9	1.8	2.0	10	2.0	3.0	3.1
Wall friction angle 2B (622 Pa) (°)	10	25.2	28.4	38.1	9	23.8	25.8	37.1	2	19.0	22.9	26.8	24	12.8	21.1	32.2
Wall friction angle 2B-EP (622 Pa) (°)	5	26.2	28.3	40.7	7	12.7	24.0	30.2	7	18.1	24.9	32.8	14	8.3	13.4	32.7

10th, 50th, and 90th are percentiles of the population.

on the 2B-EP surface than on the 2B finish) and in these instances a more intimate contact or a specific interaction between the powder and substrate is postulated to explain the results.

Comparison of Material Types

The physical property and wall friction data for each of the different material types are summarized in Table 3. Trends in the data for excipients, placebo blends, placebo granulations, and microspheres/dispersion could not be reliably determined because of their small sample populations, but they are reported for completeness.

Close inspection of the data reveals that APIs consistently recorded the highest median WFAs, active granulations the lowest, and active blends the second lowest. The excipient data were consistent with the data previously reported in the literature.^{8–11,14–16} Based on the wall friction ranking scale reported by Stanley-Wood,⁴ the majority of APIs can be rated as poor flowing materials ($\phi' > 30$), whereas most excipients and active blends can be rated as easy flowing powders ($\phi' < 20$).

The tendency of the APIs to be more frictional than the other materials is of interest because it suggests that the manufacturing equipment used to handle APIs may need to be designed and constructed differently from that used for blends and granulations. As the API is blended with excipients (often including lubricants)

and then further processed (e.g., by granulation), the friction with the manufacturing equipment is significantly reduced. Thus, the drug product manufacturing process is imparting improved materials handling properties in the majority of cases. This is one of the main justifications for the adoption of the drug product formulation and processing steps, but to our knowledge this is the first time that a substantial body of wall friction data has been analyzed in support of this premise. Consistent with the previous “all materials” analysis, the WFAs measured on the 2B stainless steel for both active blends and active granulations were usually greater than those measured on the 2B-EP surface.

Impact of API Loading

Next, the properties of the blend and granulation samples with a known level of API were analyzed and compared (Tables 4–6). Even though the size of these subsets of materials was significantly smaller, their numbers were still sufficient for a meaningful analysis. The data presented in Table 4 allow a slightly more detailed comparison of the active blend and active granulation properties. Most significantly, when compared to the active granulations the median property values for the active blends showed a marked trend toward a lower API content (5.5% vs. 12.0%) yet they still exhibited significantly higher WFAs. The densities and particle size statistics for the active blends and active granulations were

Table 4
Properties of Blends and Granulations Containing Known Amounts of API

Variable	Active Blends				Active Granulations			
	N	10th	50th	90th	N	10th	50th	90th
API load (%)	15	1.06	5.50	62.40	34	2.50	12.02	53.63
Bulk density (g/cm ³)	15	0.48	0.56	0.67	34	0.44	0.57	0.64
Tapped density (g/cm ³)	4	0.75	0.80	0.98	17	0.62	0.71	0.85
Particle size (10th percentile) (μm)	11	10.1	17.3	24.4	24	7.0	24.3	72.1
Particle size (50th percentile) (μm)	11	97.6	148.0	222.0	24	62.9	113.6	393.8
Particle size (90th percentile) (μm)	11	217.8	428.5	635.7	24	275.4	514.1	1192
Span (Bmid80)	11	2.1	2.7	3.7	24	2.0	3.1	6.4
Wall friction angle 2B (622 Pa) (°)	15	16.9	21.7	26.2	29	10.8	17.0	24.9
Wall friction angle 2B-EP (622 Pa) (°)	15	7.6	18.9	27.3	29	5.4	11.3	21.9

10th, 50th, and 90th are percentiles of the population.

Table 5

Properties of Blends Grouped According to the Amount of API

Variable	Low API <10%				Medium API 10% to <30%				High API ≥30%				All API Loads			
	N	10th	50th	90th	N	10th	50th	90th	N	10th	50th	90th	N	10th	50th	90th
API load (%)	8	0.18	2.85	5.15	2	20.00	20.00	20.00	5	35.52	43.49	75.00	15	1.06	5.50	62.40
Bulk density (g/cm ³)	8	0.48	0.58	0.69	2	0.54	0.55	0.56	5	0.49	0.54	0.60	15	0.48	0.56	0.67
Tapped density (g/cm ³)	1	1.05	1.05	1.05	0				3	0.74	0.78	0.81	4	0.75	0.80	0.98
Particle size (10th percentile) (μm)	4	17.4	17.7	24.5	2	24.1	24.3	24.4	5	7.9	11.3	13.8	11	10.1	17.3	24.4
Particle size (50th percentile) (μm)	4	102.4	130.8	229.9	2	149.4	150.8	152.2	5	67.9	131.0	209.6	11	97.6	148.0	222.0
Particle size (90th percentile) (μm)	4	240.8	329.2	457.1	2	449.2	532.1	615.0	5	228.8	477.0	701.8	11	217.8	428.5	635.7
Span (Bmid80)	4	1.9	2.2	2.3	2	2.8	3.4	3.9	5	2.9	3.6	3.7	11	2.1	2.7	3.7
Wall friction angle 2B (622 Pa) (°)	8	19.0	23.2	28.1	2	19.5	20.8	22.2	5	16.1	20.4	22.4	15	16.9	21.7	26.2
Wall friction angle 2B-EP (622 Pa) (°)	8	7.1	18.9	30.9	2	20.0	21.5	23.0	5	9.8	15.7	20.7	15	7.6	18.9	27.3

10th, 50th, and 90th are percentiles of the population.

Table 6

Properties of Granulations Grouped According to the Amount of API

Variable	Low API <10%				Medium API 10% to <30%				High API ≥30%				All API Loads			
	N	10th	50th	90th	N	10th	50th	90th	N	10th	50th	90th	N	10th	50th	90th
API load (%)	15	1.25	3.60	5.90	10	11.25	20.00	23.05	9	31.05	52.29	69.28	34	2.50	12.02	53.63
Bulk density (g/cm ³)	15	0.42	0.57	0.65	10	0.51	0.57	0.64	9	0.47	0.51	0.59	34	0.44	0.57	0.64
Tapped density (g/cm ³)	10	0.58	0.67	0.83	6	0.70	0.72	0.81	1	0.86	0.86	0.86	17	0.62	0.71	0.85
Particle size (10th percentile) (μm)	12	19.1	30.9	73.4	6	6.7	7.9	57.4	6	8.6	18.1	29.8	24	7.0	24.3	72.1
Particle size (50th percentile) (μm)	12	111.2	184.0	507.7	6	55.5	92.1	302.5	6	60.7	75.6	179.3	24	62.9	113.6	393.8
Particle size (90th percentile) (μm)	12	280.0	502.5	1273	6	284.0	507.5	812.0	6	293.7	524.6	812.0	24	275.4	514.1	1192
Span (Bmid80)	12	2.0	2.2	3.1	6	2.7	3.7	8.0	6	3.2	5.2	9.6	24	2.0	3.1	6.4
Wall friction angle 2B (622 Pa) (°)	14	10.3	12.6	22.4	8	15.1	23.5	28.6	7	11.6	14.9	22.2	29	10.8	17.0	24.9
Wall friction angle 2B-EP (622 Pa) (°)	14	5.1	12.2	21.6	8	6.4	19.9	22.7	7	5.2	8.3	12.9	29	5.4	11.3	21.9

10th, 50th, and 90th are percentiles of the population.

generally quite similar, although the blends had fewer coarse particles and a slightly narrower size distribution on average. These differences are expected based on the normal effects of pharmaceutical granulation processes. In Tables 5 and 6, the data for the active blends and active granulations are reanalyzed in groupings with low (<10%), medium (10% to <30%), and high (≥30%) API levels. Most of these groups had quite low numbers of samples ($N < 10$), so the trends should not be over-interpreted. In fact, no new trends could be identified from this alternate stratification of the data. Perhaps surprisingly (because of the greater wall friction of the APIs), the API level did not appear to have a pronounced impact on the wall friction values recorded *per se*, nor did it appear to impact the density or particle size of the samples in a systematic manner. As noted previously, the most marked differences were

between the active blends and granulations and between the standard and electropolished surface finishes.

Advanced Statistical Analysis

Finally, the data were examined for trends among the WFA and particle size, bulk density, and tapped density data. Bivariate plots of WFA against each of the physical properties all showed considerable scatter, with a typical plot shown in Figure 4. The correlation coefficient (R^2) values for the best fit curves on these plots were all less than 0.50. The strongest relationship was observed between

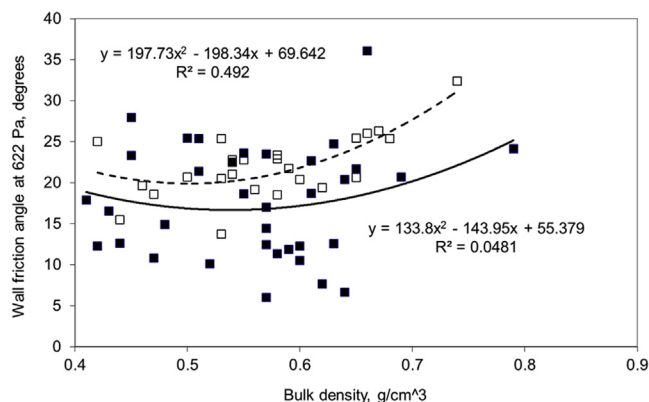


Figure 4. Relationship between wall friction angle (on stainless steel 2B surfaces) and bulk density. Closed symbols, active granulations; open symbols, active blends.

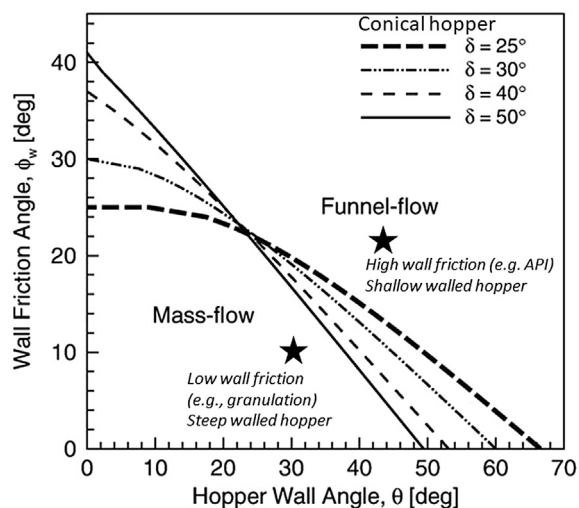


Figure 5. Example of a Jenike "design chart" used for determining the flow mode during discharge of powders (δ = powder internal angle of friction). Adapted from Ketterhagen et al.²⁷

Table 7
Effective Angle of Internal Friction Data for Different Material Types

Material Type	Number of Samples (N)	10th	50th	90th	Minimum	Maximum	Mean
All materials	123	40.0	46.0	59.6	35.0	74.0	47.8
Active blends	25	40.0	46.0	61.2	38.0	70.0	48.7
Active granulations	40	40.0	47.0	52.0	39.0	55.0	46.8
Active pharmaceutical ingredients	10	45.8	52.0	65.9	35.0	74.0	54.6

10th, 50th, and 90th are percentiles of the population.

the bulk density of the active blends and the wall friction against the 2B finish ($R^2 = 0.492$), while the corresponding plot for granulations was extremely weak. The lack of any significant relationship between the wall friction values and the physical properties of the samples is consistent with previous reports in the literature.¹ Multivariate method (including partial least squares) approaches were used to analyze the data and to look for correlations between combinations of the measured physical properties. Despite extensive data review and stratification no definitive trends could be discerned using these techniques.

Practical Significance

Example of Flow Mode Calculation

As mentioned in the Background section, the most common use of wall friction data is to predict the flow mode (“mass” or “funnel” flow) during the gravity discharge of powders from bins and hoppers used in manufacturing processes. This is important because funnel flow patterns can result in a variety of processing and product quality problems, including segregation, erratic flow rates, material hang-up, poor content uniformity, and low product yields.²⁵ Usually the so-called “design charts” developed by Jenike²⁶ are used for this purpose (Fig. 5).

Separate design charts exist for hoppers of different designs (conical, slot opening, etc.), and the first step is to select the correct design chart for the hopper or bin of interest. Next, the appropriate boundary condition is identified according to the internal friction (δ) of the powder of interest. We have summarized the effective angle of internal friction data for the pharmaceutical powder samples considered in this work and the results are presented in Table 7. Finally, the wall angle of the hopper of interest is located on the x-axis of the design chart and the wall friction value is located on the y-axis, and the coordinate comprising these 2 points is plotted on the graph. The location of this point relative to the mass/funnel flow boundary line indicates what type of flow mode should be expected.

Based on the results of this work, we can use the design charts to predict the likely flow behavior of different pharmaceutical materials in common hoppers and bins (see star symbols on Fig. 5). For common “off-the-shelf” conical bins and hoppers with valley angles of between 30° and 45° from vertical, it is clear that most pharmaceutical blends and APIs (typical WFA >20°) would be expected to exhibit funnel flow behavior. For active granulations (typical WFA <20°), the significantly lower friction with the surface of the equipment should lead to mass flow in a much larger number of instances. From this simple example, it can be readily appreciated that wall friction data can have a profound impact on the manufacturing performance of solid pharmaceutical products. Hence, it is critical to have an appreciation for the range of normal product properties that may be encountered during routine operations. The results of this work should help to provide this greater material property understanding and the appropriate context to interpret any additional data that are generated in the future for new pharmaceutical products.

Conclusions

Wall friction data for over 100 pharmaceutical powders have been analyzed and the major trends in the data identified. This analysis significantly improves our understanding of how pharmaceutical active ingredients, blends, and granulations will interact with vessel walls during powder processing operations. Overall, the wall friction data are consistent with the small amount of data previously reported in the literature. Blends tend to exhibit larger WFAs than granulations, and this means that blends will usually require hoppers or bins with steeper walls than granulations to achieve mass flow. Lower WFAs were often observed with smoother surfaces, and thus the wall surface finish should be a significant consideration when designing vessels for handling these types of materials. Using the data and trends presented in this work, it should be possible to more intelligently select, design, and build hoppers and bins for the handling of pharmaceutical powders, and thus avoid the problems associated with funnel flow patterns, such as segregation, yield losses, and erratic flow rates. Developers and manufacturers should take special care when choosing from existing hoppers or bins (avoiding shallow slopes and unpolished surfaces) and should also include lubricants and flow aids in formulations when warranted.

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